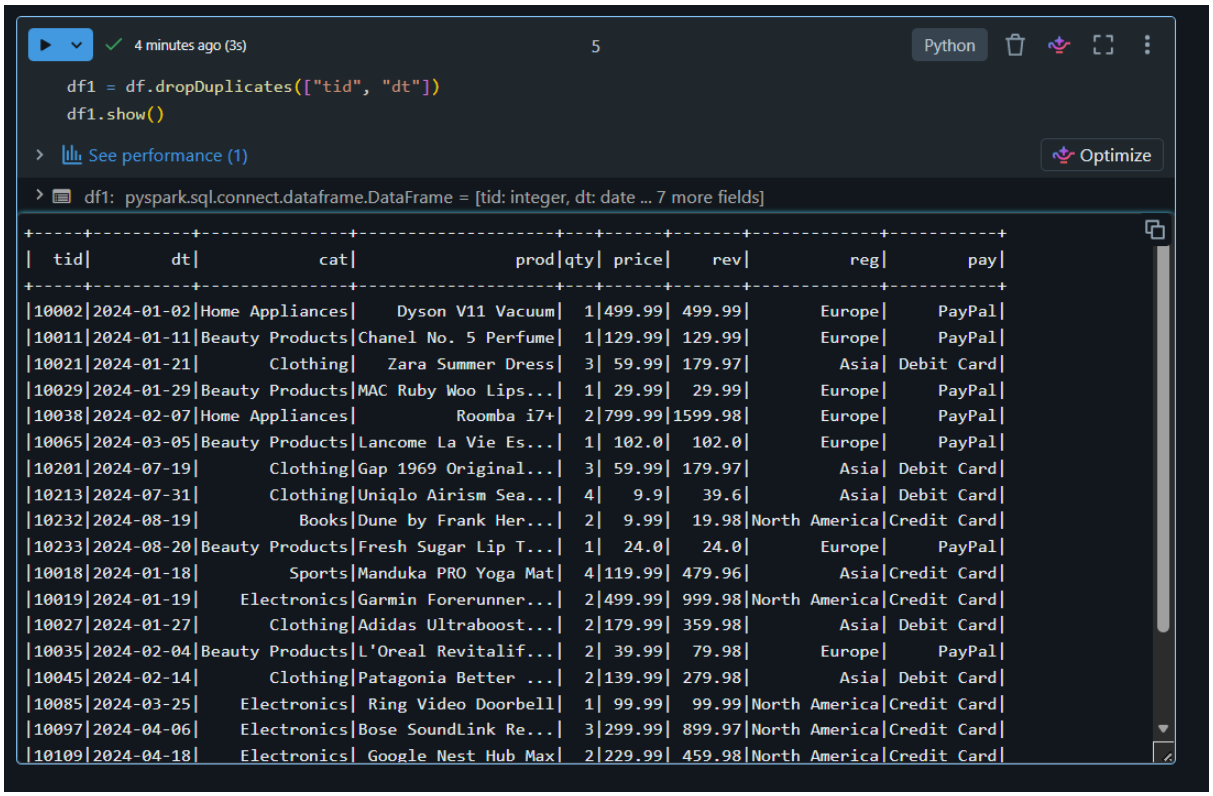
Q1: What are the key limitations of traditional MapReduce that make Spark a preferred choice for modern big data processing?

MapReduce is slow because it stores data on the disk after every step. This increases processing time. Spark is faster because it keeps data in memory (RAM), reducing disk usage. It also supports machine learning, SQL, and real-time processing, making it a better choice for big data.

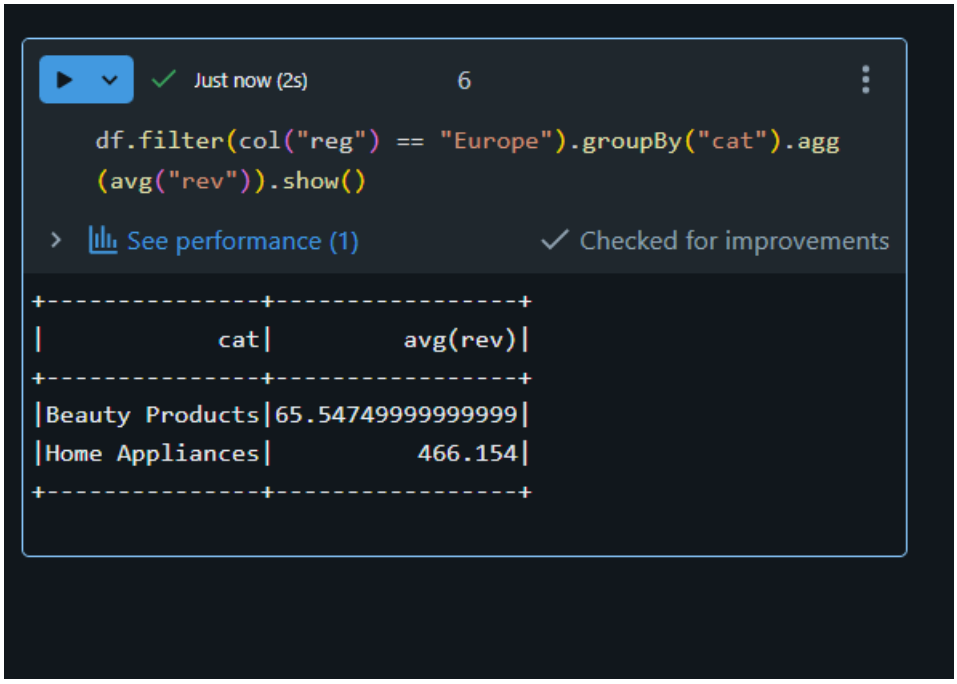
Q2: Explain how Spark uses In-Memory Computing to speed up iterative machine learning algorithms compared to disk-based systems.

Spark stores data in memory (RAM) instead of reading it from the disk every time. This makes repeated tasks like machine learning much faster because the same data can be reused without loading it again and again.

Q3: Write a code snippet to remove all duplicate rows from a DataFrame based on user_id and transaction_date.



Q4: Given a DataFrame df_sales, filter rows where the region is 'West' and group by product_category to find the average sale_amount.



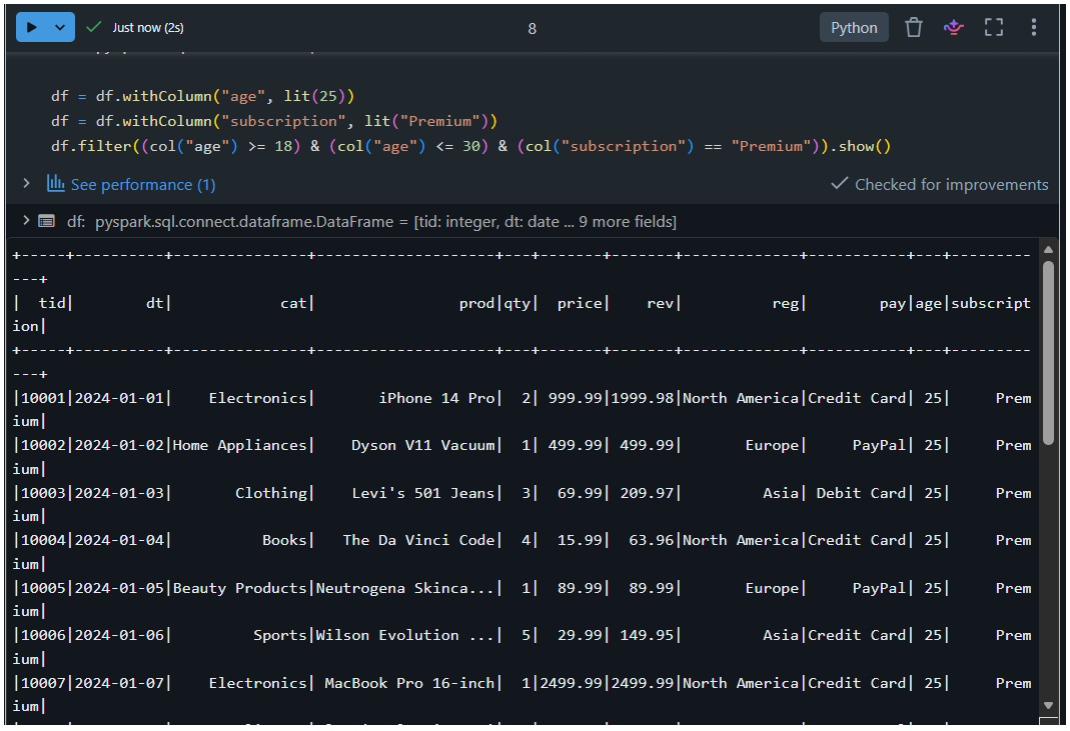
Q6: Write a query to find the total count of records for each city in a DataFrame, but only for cities where the count is greater than 100.

No rows returned because no region has more than 100 records.

Q7: How does the immutability of Spark DataFrames affect how you perform "data cleaning" steps like dropping columns or renaming them?

Spark DataFrames are **immutable**, which means the original DataFrame cannot be changed. When we drop or rename columns, Spark creates a **new DataFrame** instead of modifying the existing one. This helps keep the original data safe.

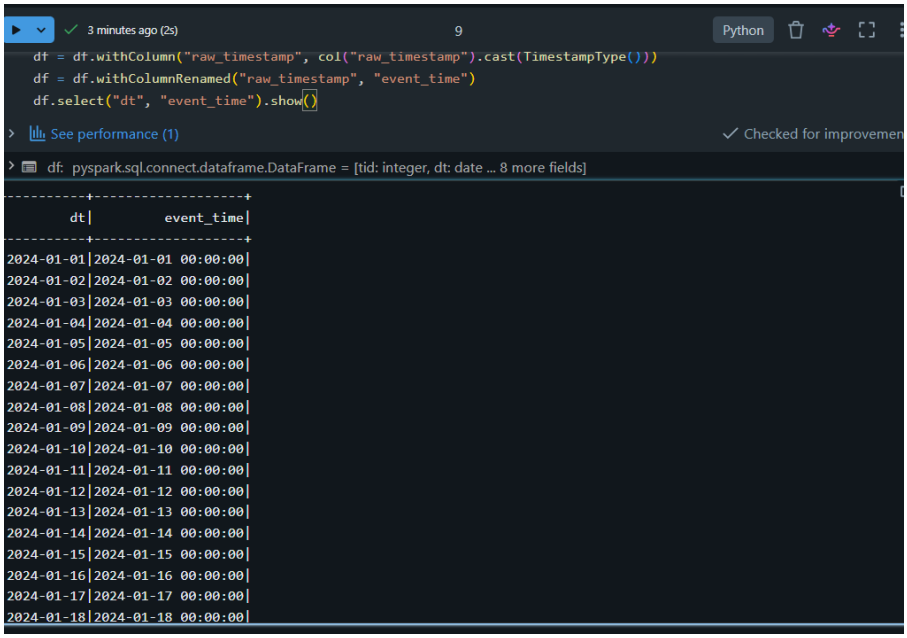
Q8: Write a Spark command to filter a dataset for rows where the age is between 18 and 30 (inclusive) and the subscription is 'Premium'.



Q9: When cleaning a dataset, why is it often better to handle null values before performing mathematical aggregations like sum() or avg()?

It is better to handle null values first because they can affect the accuracy of calculations like sum() and avg(). Filling or removing null values helps produce correct and reliable results.

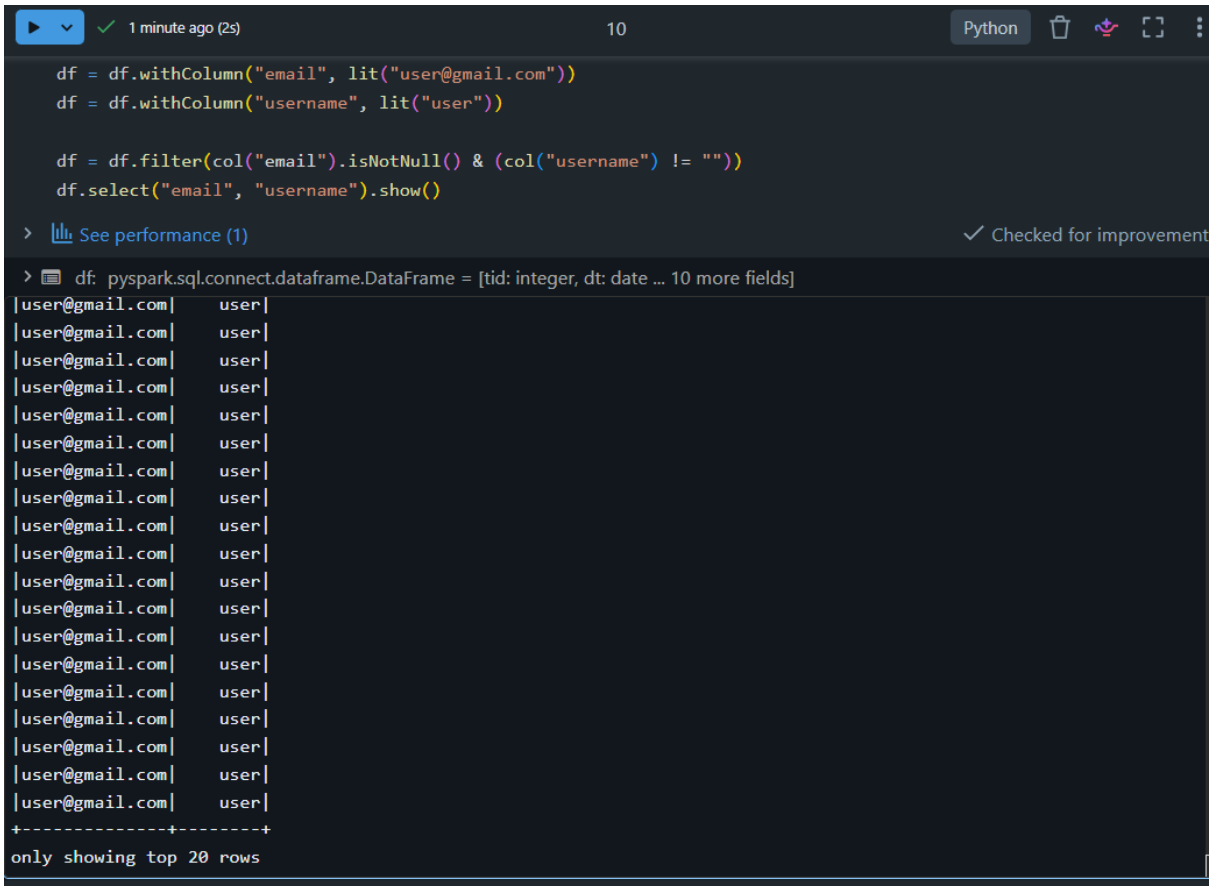
Q10: Write the code to revise a column named raw_timestamp by casting it to a TimestampType and renaming it to event_time.



Q11: Explain the "Shuffle" process that occurs during a grouping operation. Why is it considered a wide transformation?

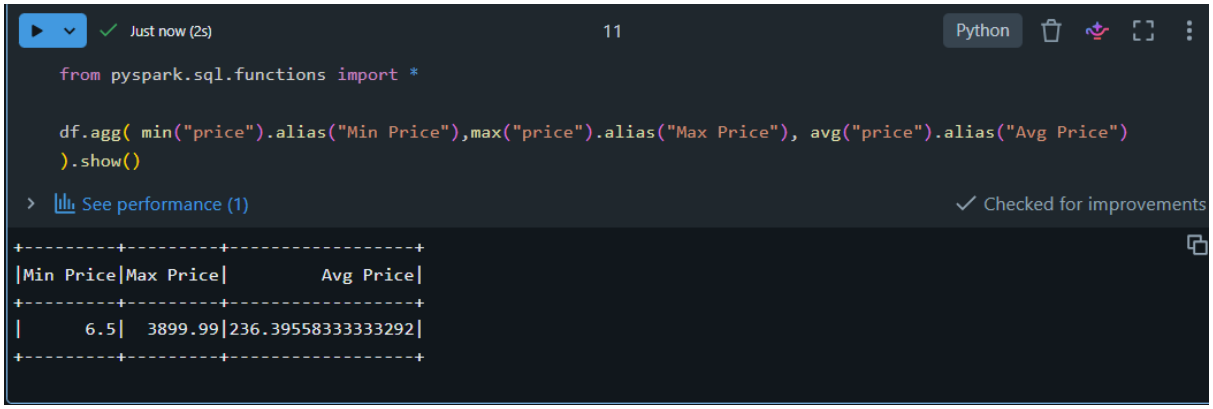
Shuffle is the process of moving data between different partitions while performing operations like groupBy(). It is called a **wide transformation** because data from multiple partitions is exchanged, which makes the operation slower than narrow transformations.

Q12: Write a code snippet that identifies and removes rows where the email column contains null values OR the username is an empty string.



All rows contain valid email and username values, so no rows were removed.

Q13: How do you use the .agg() function to calculate multiple statistics at once, such as the min, max, and mean of the price column?



Q14: In the context of cleaning a dataset, what is the risk of using inferSchema=true when your source data contains messy or inconsistent date formats?

If the date formats are inconsistent, inferSchema=true may detect the wrong data type or treat dates as strings. This can cause errors or incorrect results during data processing.

Q15: Write a final processing pipeline that:

1. **Filters out duplicates.**
2. **Fills null prices with 0.**
3. **Groups by store_id to calculate total revenue.**